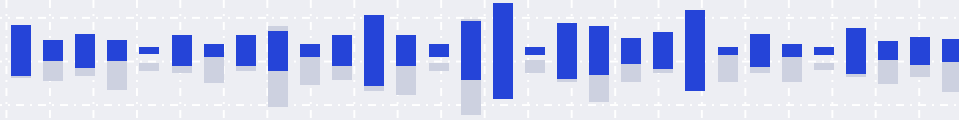


Unmet Basic Needs and Escalated Care Journeys

Arshia Singla, Sally Han, Angela Liu, Yuqi Liu

01

Problem Statement



Are patients with unmet basic needs more likely to experience escalated care journeys, defined by emergency visits or hospital admissions within a diagnosis?





How we approached the problem

Patient Journeys

Patient journeys were grouped by **patient + diagnosis value**

Escalation

A journey was marked escalated if it included:

- an ED visit or
- a hospital admission

Unmet basic needs

We engineered four interpretable risk domains:

- Housing Stability
- Transportation Needs
- Financial Resource Strain
- Food Insecurity

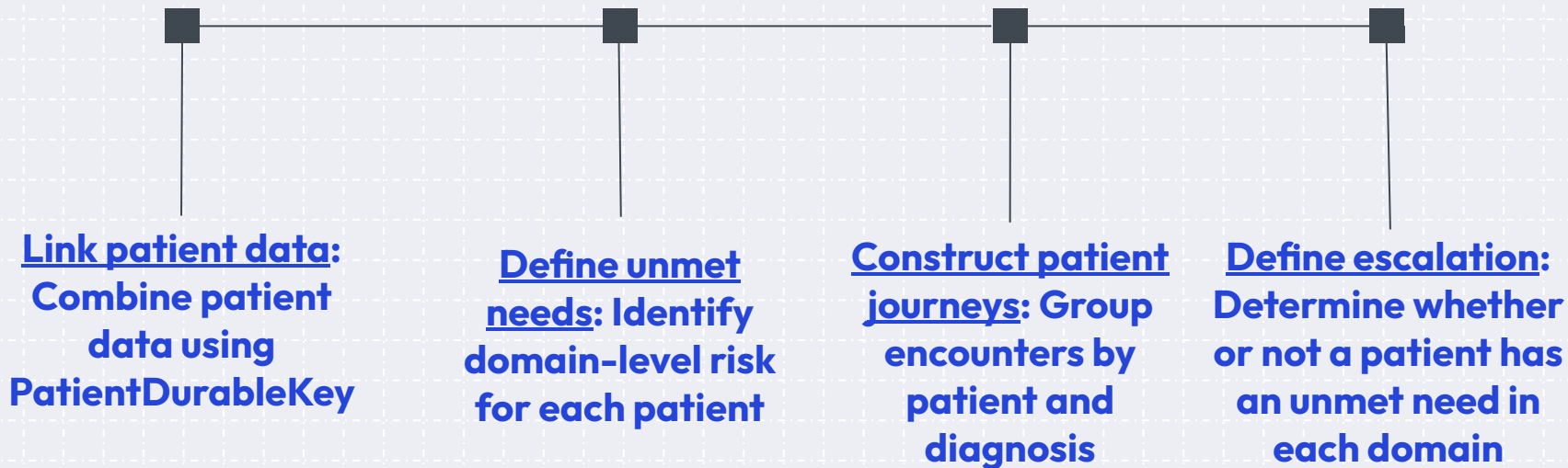


02

Methods



Methods we used to analyse the data



Modeling: logistic regression with demographic adjustments

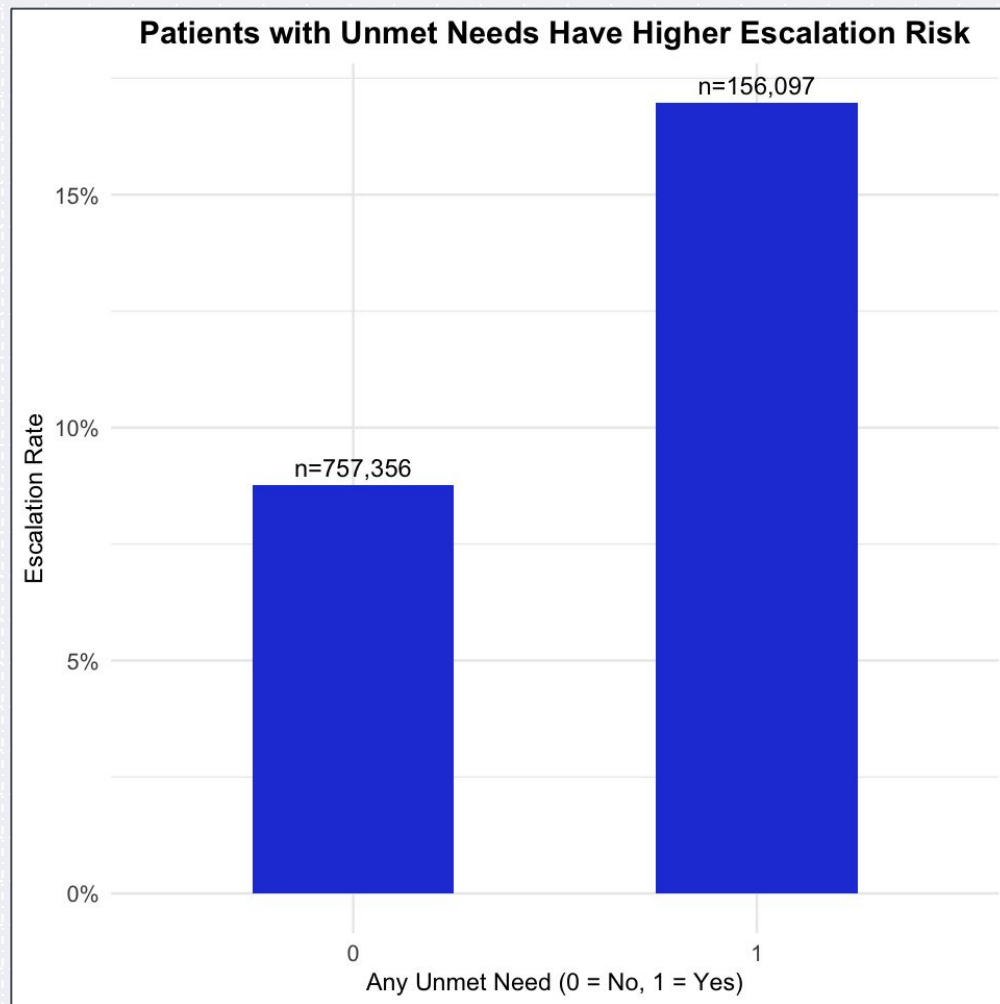
03

Insights



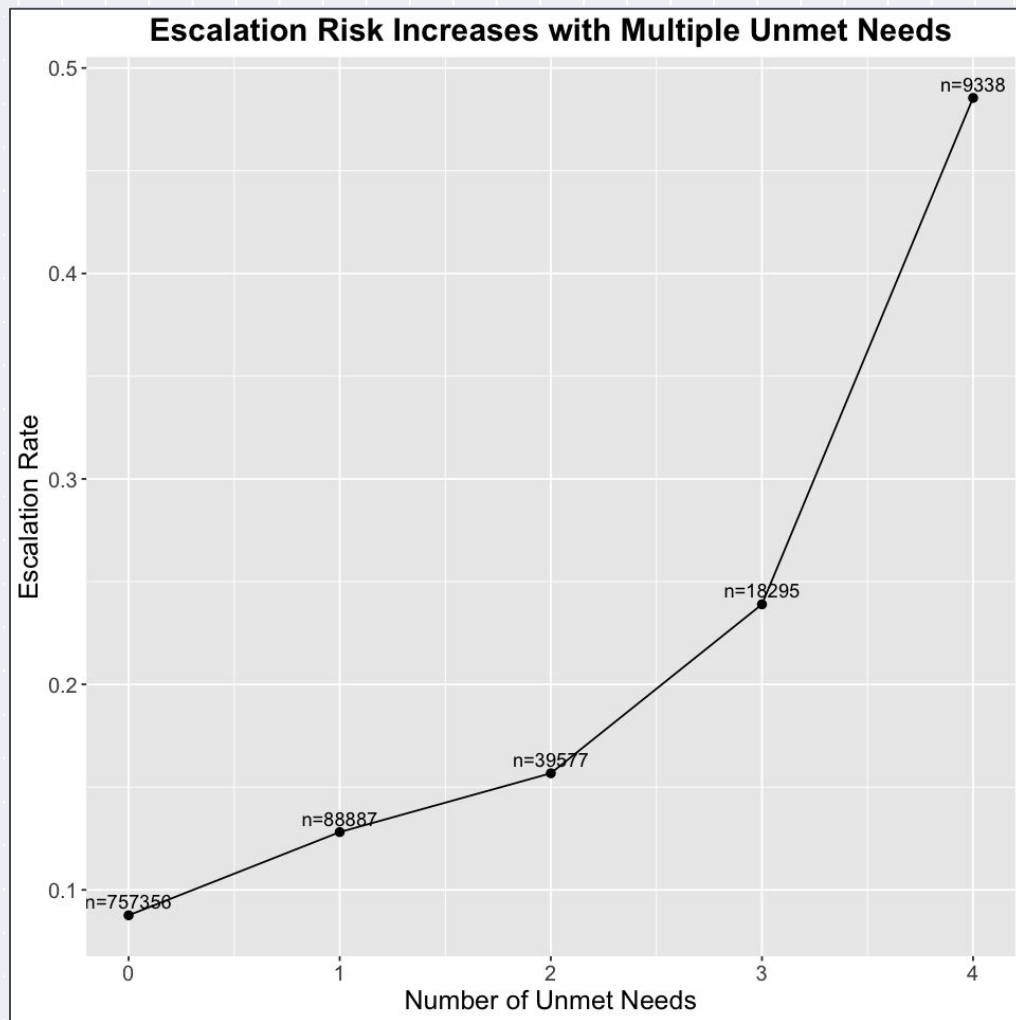
Patients with any unmet need face higher escalation risk

Escalation is nearly twice as high for patients with at least one unmet need.



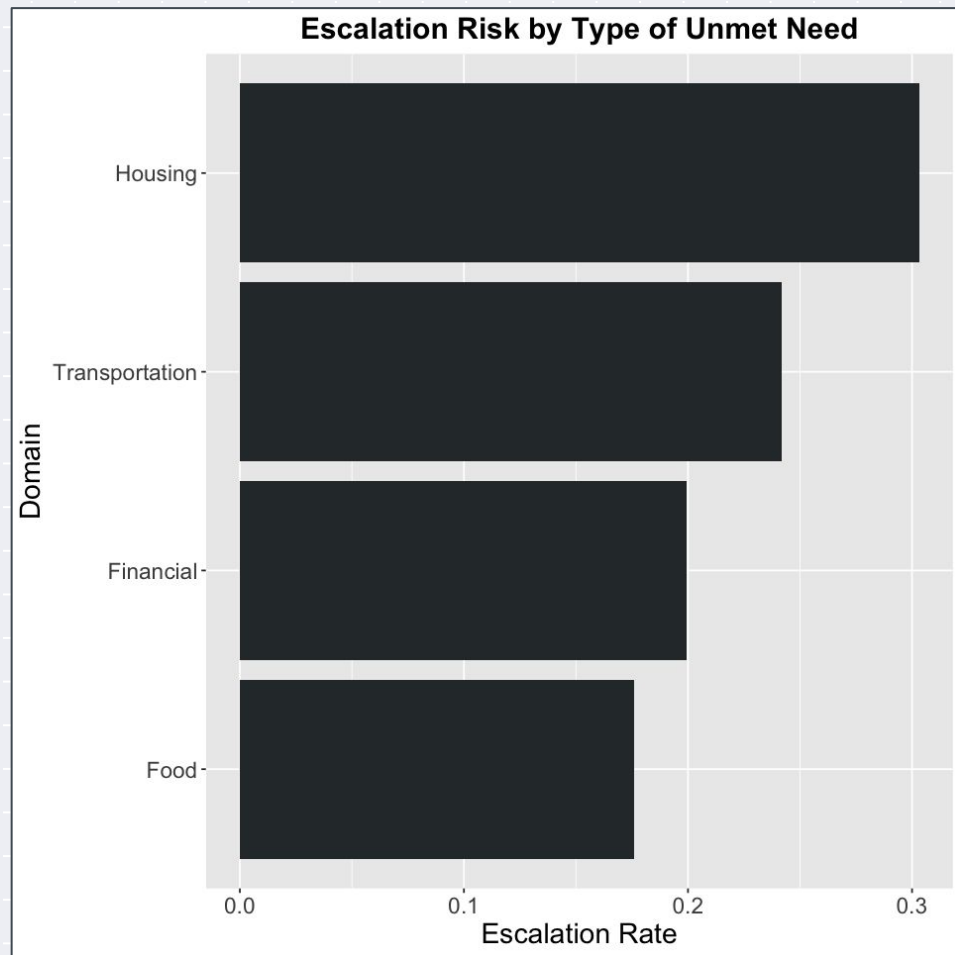
Escalation rises sharply as vulnerabilities accumulate

- 0 unmet needs: 8.8%
- 1 unmet need: 12.1%
- 2 unmet needs: 15.7%
- 3 unmet needs: 23.9%
- 4 unmet needs: 48.5%



Some unmet needs are stronger signals than others

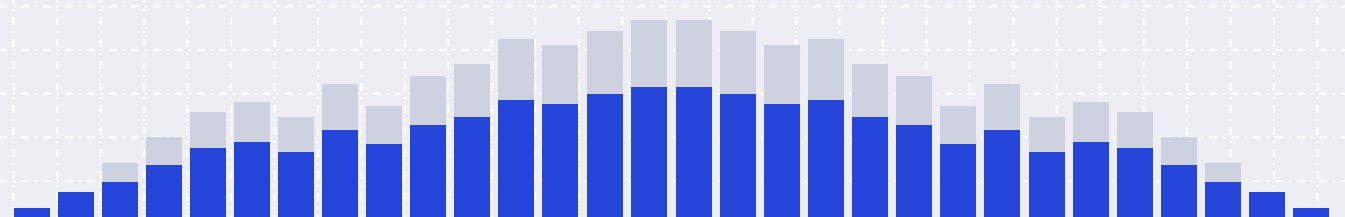
This shows that not all domains are equally associated with escalation.

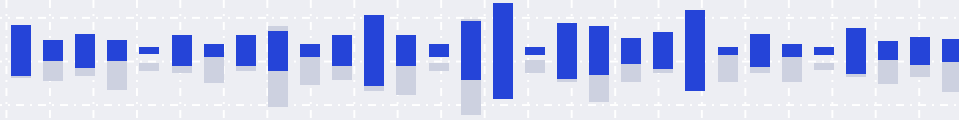




**Each additional unmet need is
associated with a ~53.6%
increase in the odds of
escalation**

Adjusted for age group, sex assigned at birth, and race





1. Unmet basic needs are a **strongly linked** to escalated care journeys.
2. Risk **increases sharply** as patients accumulate more needs.
3. This **pattern holds** even after demographic adjustment.



04

Limitations





01

Loss of detail in responses

Unspecified responses were omitted due to data limitations.

02

No causal interpretation

Observational data identify associations, not causality

03

Unobserved factors

We focused on key variables, but other unmeasured factors may also influence escalation.



05

Recommendations





**Connecting those with unmet
needs to community
resources**

**Partnering with local
organizations that improve
Social Determinants of Health**

**Engaging with patients with lived experiences of unmet needs to
design solutions**



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**Thank you for
listening!**

A decorative bar chart at the bottom of the slide, consisting of numerous vertical bars of varying heights in a dark blue color, set against a background of a light grey dashed grid.